ETH zürich

Wireless Networking and Mobile Computing

Emerging Technologies
IEEE 802.11 Wi-Fi - MAC Layer

Stefan Mangold
stefan.mangold@inf.ethz.ch

Course Outline

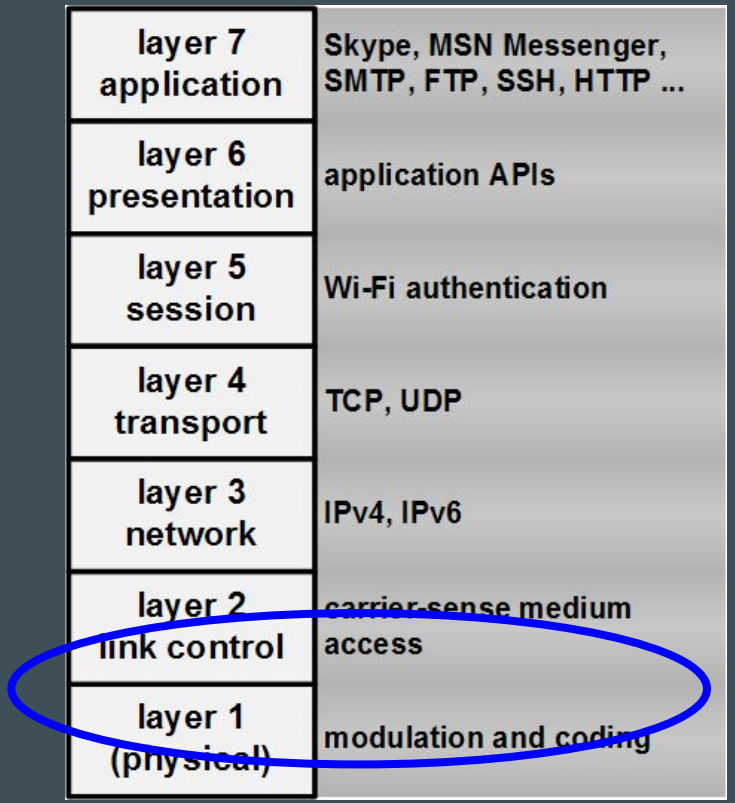
Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

ISO/OSI Reference Model - Layer 1 and Layer 2

- ISO: International Organization for Standardization
- OSI: Open Systems Interconnection



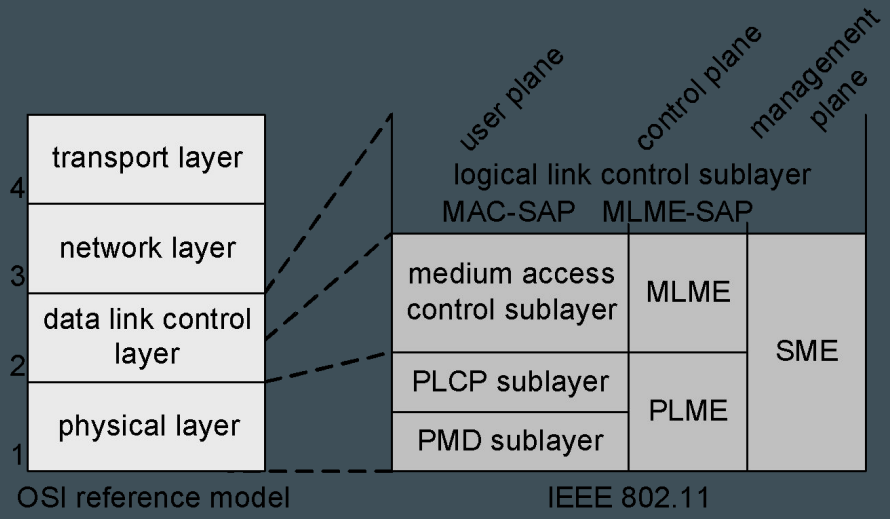
IEEE 802.11™ Overview

- Root document: IEEE 802.11 approved 1997
  - Amendments: extensions & refinements of IEEE 802.11
  - IEEE Std 802.11-2007: replacing IEEE 802.11 a/b/d/g/h...
  - IEEE Std 802.11™ -2020 was published in February, 2021
- Wi-Fi Alliance
  - Global, non-profit organization
  - Goal: Driving the adoption of a single worldwide standard
  - Test plans, interoperability & certification
  - “Wi-Fi certified” is a trademarked brand



IEEE 802.11 Reference Model

- Layer 1: PHY transmission schemes
- Layer 2: MAC protocols, MAC Layer Management Entity (MLME)
- Layer 1 & 2: Station Management Entity (SME)
- Interface to higher layers: Service Access Points (SAPs)



IEEE 802.11 List of Extensions (1/2)

document	content	status
IEEE 802.11	1 Mb/s and 2 Mb/s, @ 2.4 GHz, RF and IR	1997
IEEE 802.11a	54 Mb/s @ 5 GHz	1999
IEEE 802.11b	Enhancements to 802.11 to support 5.5 and 11 Mb/s	1999
IEEE 802.11c	Bridge operation procedures; included in IEEE 802.1D	2001
IEEE 802.11d	International (country-to-country) roaming extensions	2001
IEEE 802.11e	Quality-of-Service enhancements	2005
IEEE 802.11g	54 Mb/s, @ 2.4 GHz (backwards compatible with b)	2003
IEEE 802.11h	Spectrum managed 802.11a (5 GHz)	2004
IEEE 802.11i	Enhanced security	2004
IEEE 802.11j	Extensions for Japan	2004
IEEE 802.11k	Radio resource measurement enhancements	2007
IEEE 802.11m	Maintenance of the standard	always ongoing

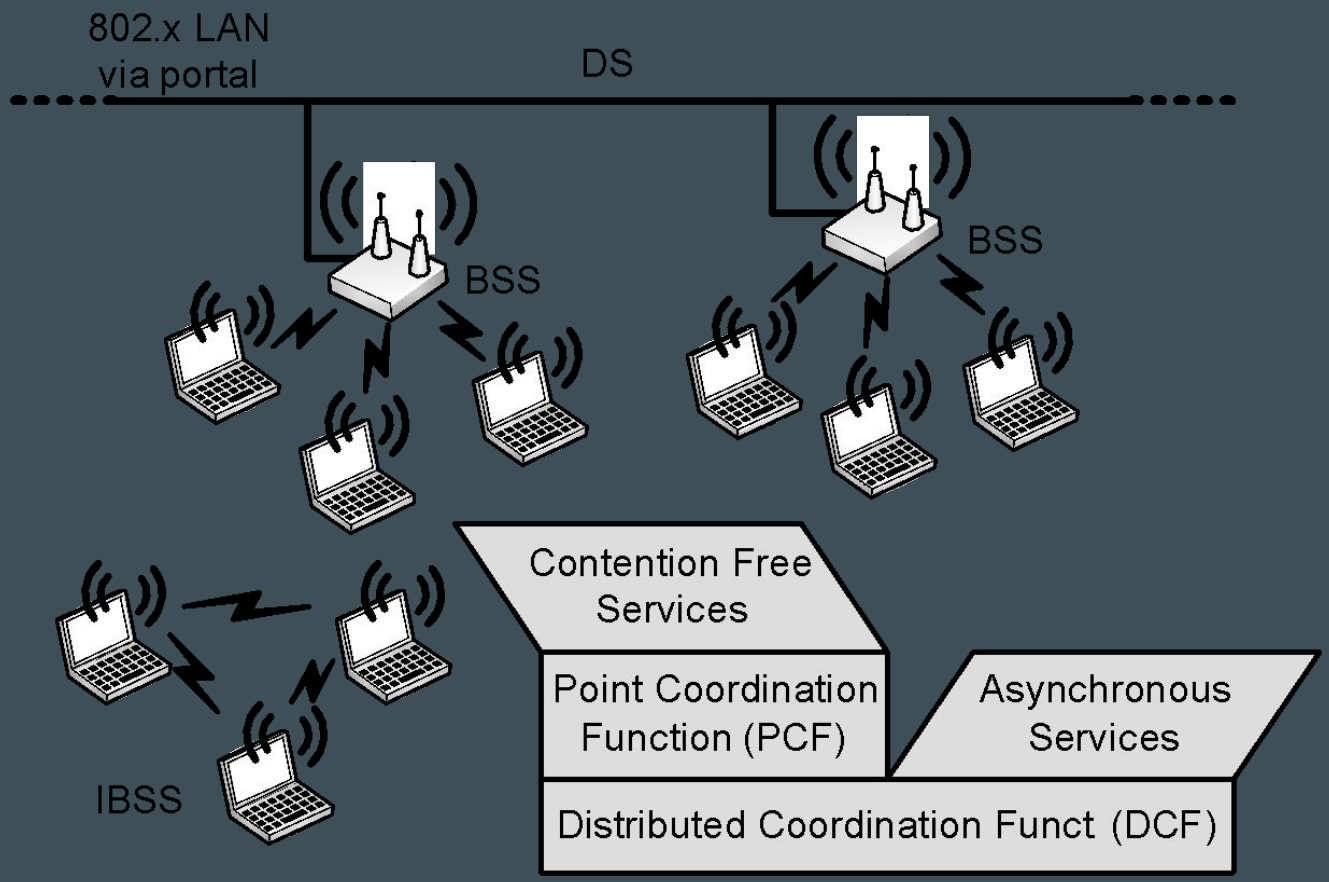
IEEE 802.11 List of Extensions (2/2)

document	content	status
IEEE 802.11n	Higher throughput improvements using MIMO	2008
IEEE 802.11p	WAVE - Wireless Access for Vehicular Environment	2009
IEEE 802.11r	Fast roaming	2007
IEEE 802.11s	ESS Extended Service Set Mesh Networking	2008
IEEE 802.11u	Interworking with non-802 networks (cellular)	2009
IEEE 802.11v	Wireless network management	2012
IEEE 802.11w	Protected Management Frames	2012
IEEE 802.11y	3650-3700 MHz Operation in U.S.	2009
IEEE 802.11z	Direct Link Setup	2010
...	... more extensions exist	...
IEEE 802.11ad	60GHz	2018, ongoing
IEEE 802.11af	TV White Space	2016
IEEE 802.11ah	Sub 1GHz, Internet of Things	2018
IEEE 802.11ax	Wi-Fi 6: all of the above, plus higher modulation, OFDMA, power mgmt	2021

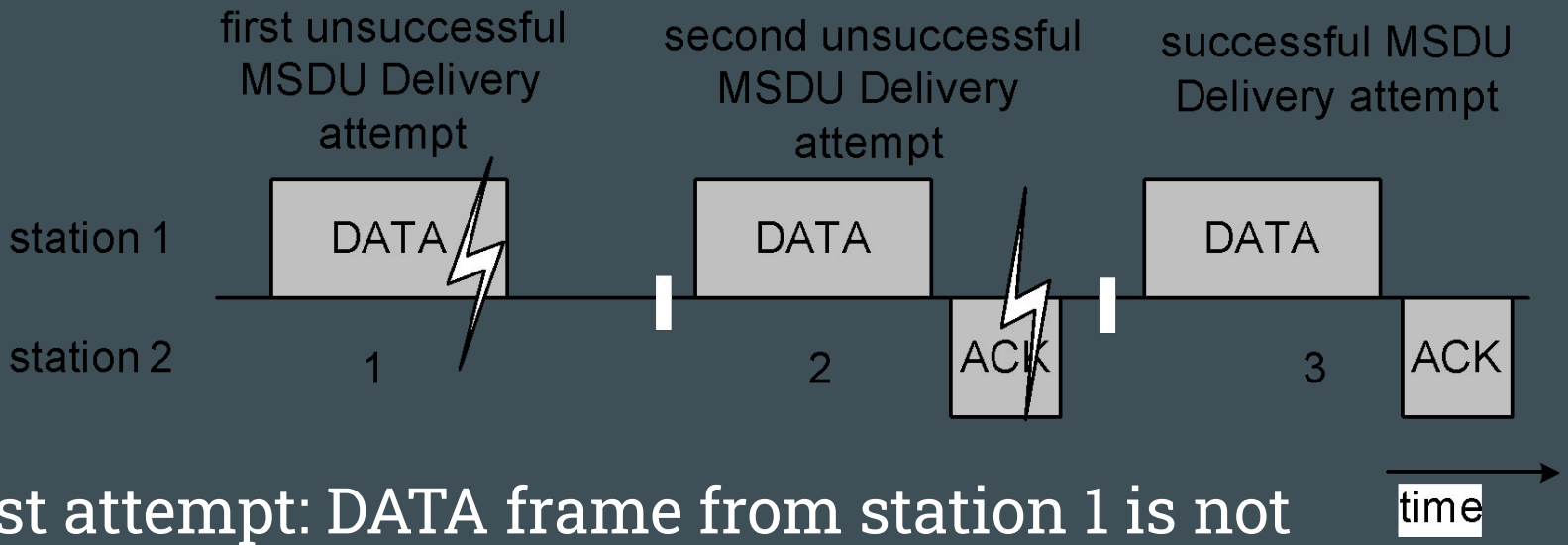
IEEE 802.11 Architecture

- BSS = Basic Service Set, with access point
- IBSS = Independent BSS, for ad hoc networks
- DS = Distribution System
- Distributed Coordination Function (DCF) = distributed control

IEEE 802.11 Architecture Illustrated

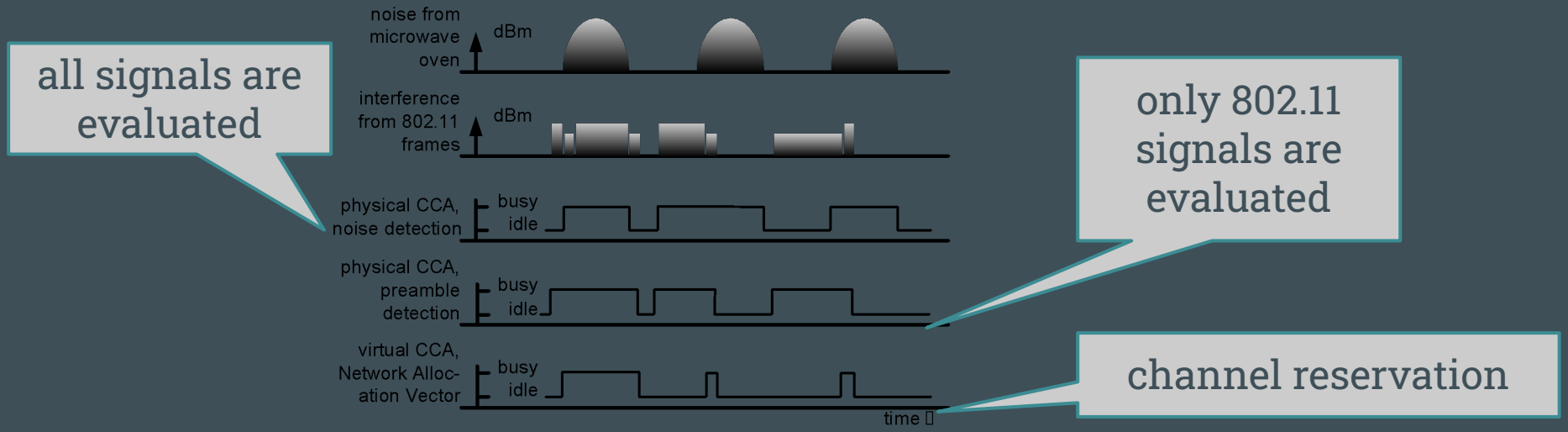


802.11 Example Frame Exchange



- First attempt: DATA frame from station 1 is not received, no ACK
- Second attempt: DATA is successfully received, ACK is sent but missed by station 1 (might be because of interference)
- Third and successful attempt: DATA and ACK are received

Carrier Sensing (”listen before talk”)

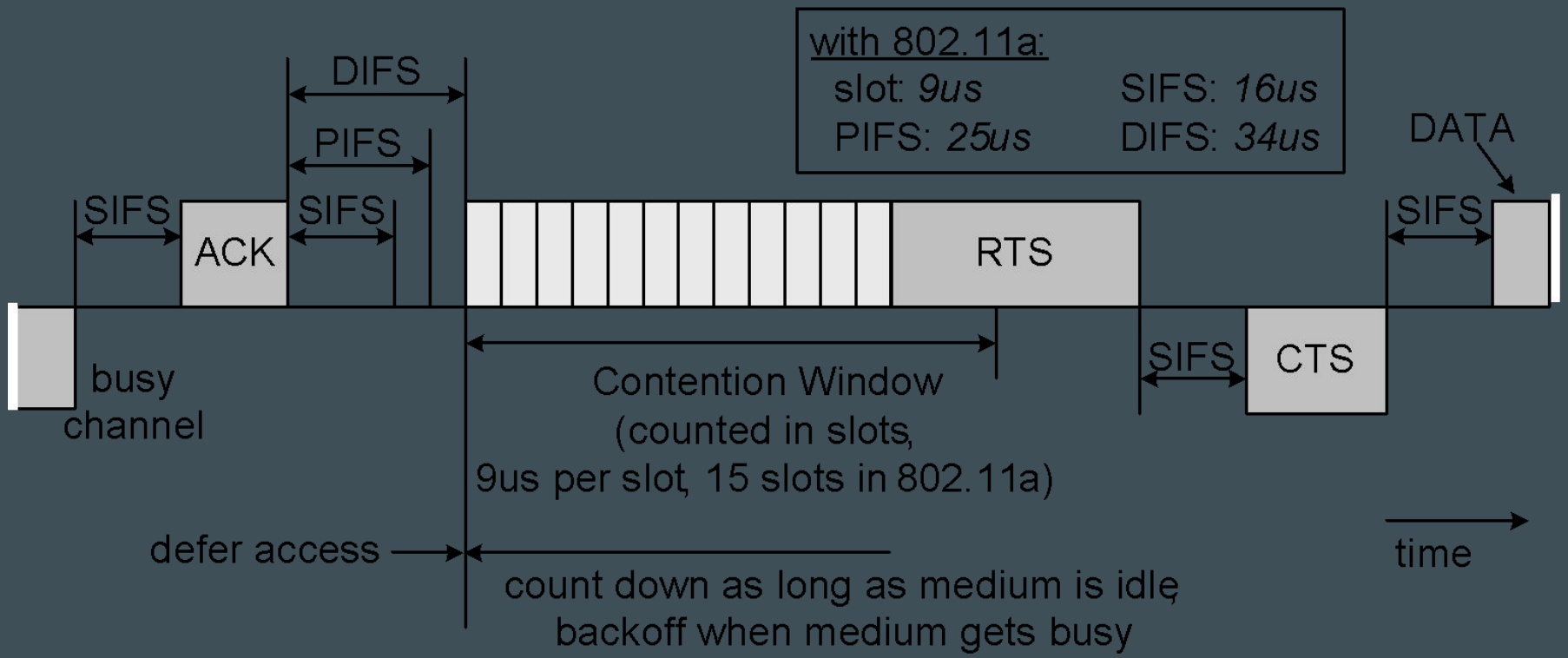


- Channel sensing function: Clear Channel Assessment (CCA)
  - Channel determined as busy when received power is larger than fixed power threshold of -82dBm
  - Network Allocation Vector (NAV): timer for reserving channel, defines duration of following frame exchange. No transmissions are allowed until timer is expired.

Distributed Coordination Function (DCF)

- Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
  - listen-before-talk principle and backoff procedure
- CCA carrier sensing is used to detect idle channel
  - CCA = Clear Channel Assessment
  - collision avoidance (not: Collision Detection)
- Packet transmission is initiated after certain idle time:
  - SIFS = Short Interframe Space (no CCA carrier sensing)
  - PIFS = PCF Interframe Space = SIFS + aSlotTime (with CCA)
  - DIFS = DCF Interframe Space = PIFS + aSlotTime (with CCA)
- SIFS is needed for transceiver turnaround

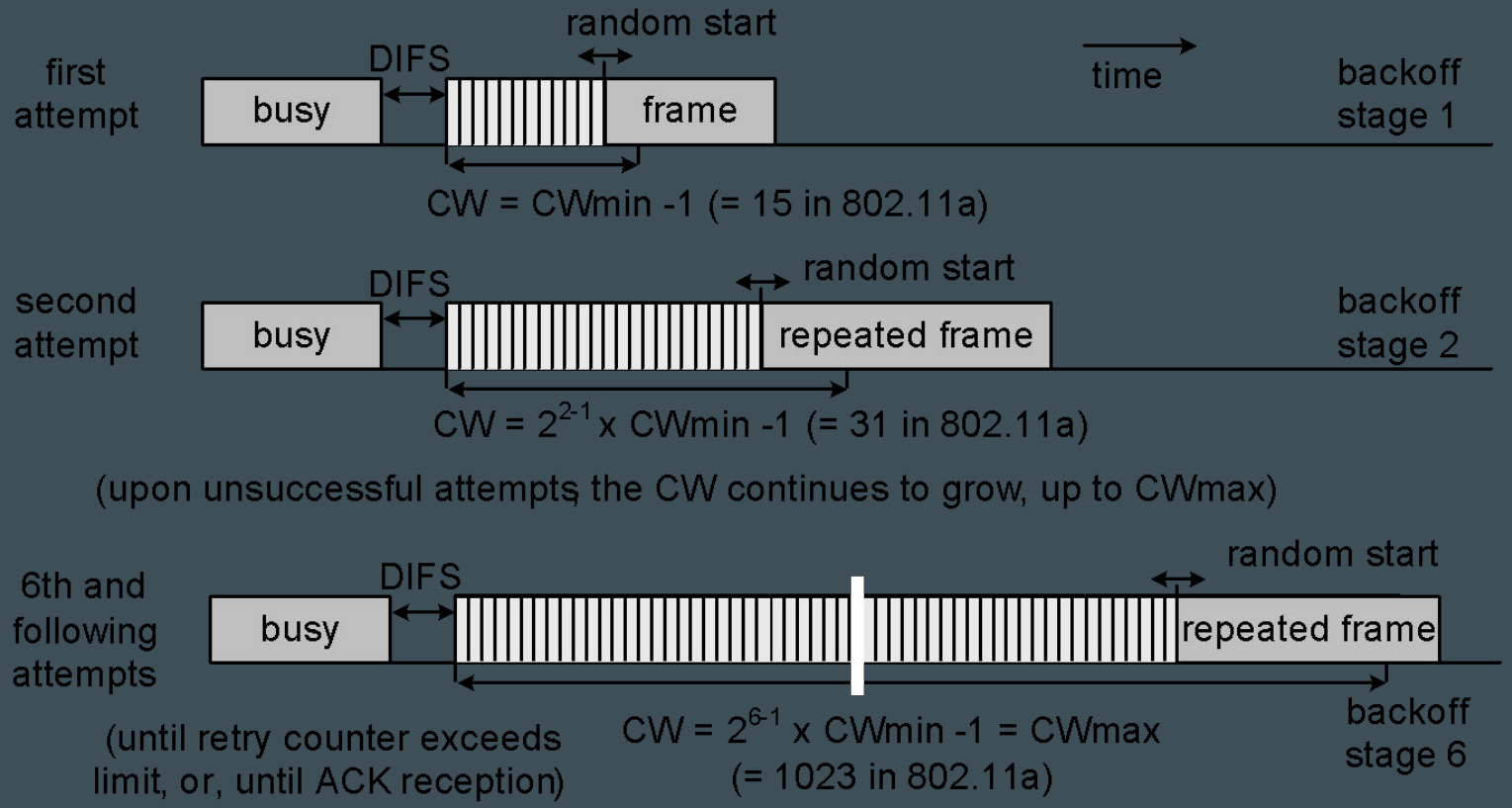
DCF Illustrated

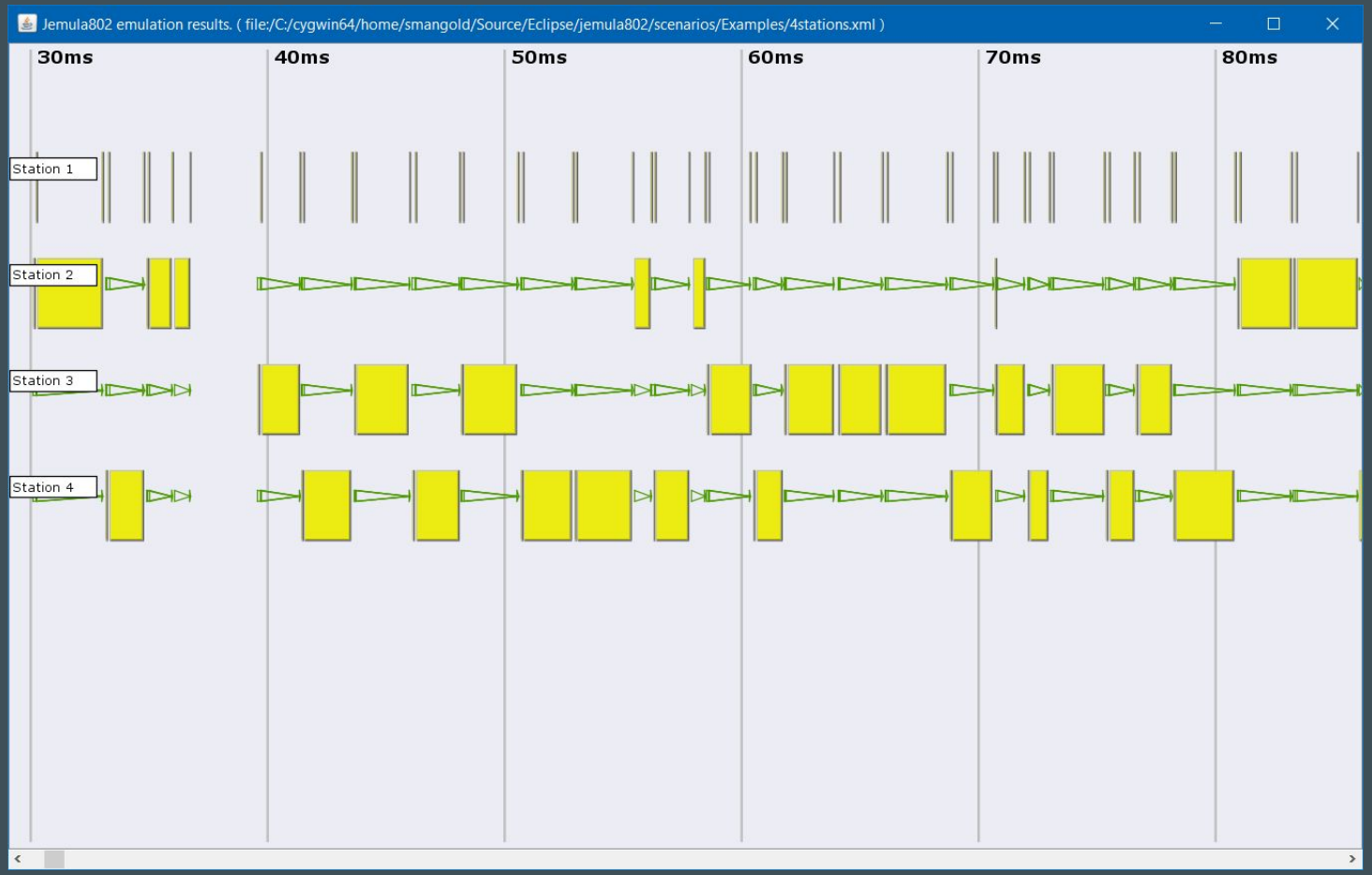


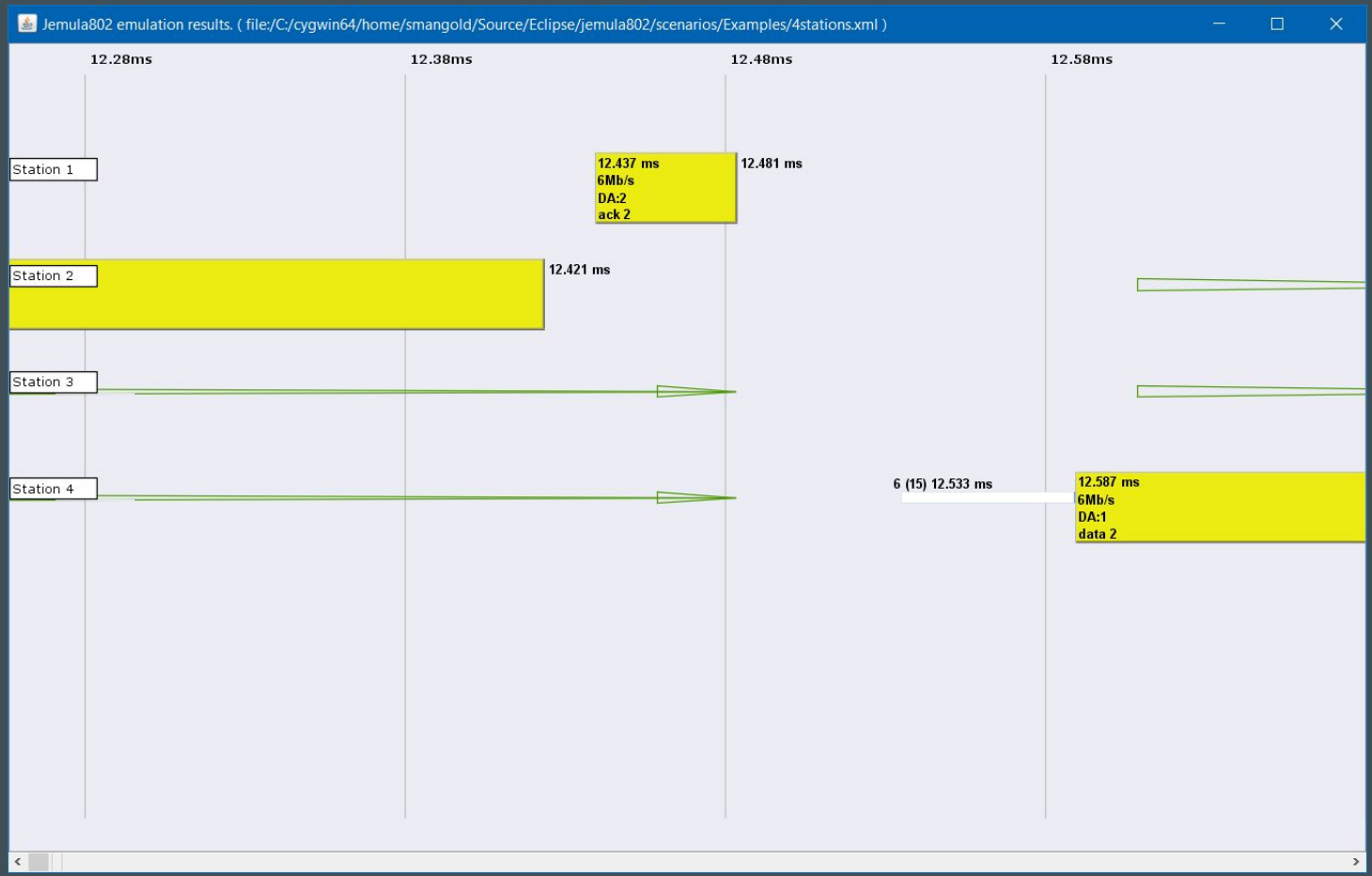
DCF Backoff Window

- Backoff is performed in case of collision
  - Random value is drawn out of contention window
- The size of the Contention Window (CW) is exponentially increased in case of repeated collisions
- Backoff procedure has two parameters
  - Minimum contention window size CWmin
  - Maximum contention window size CWmax
  - Example values for 802.11a: CWmin=16 and CWmax=1024

DCF Backoff Window Illustrated



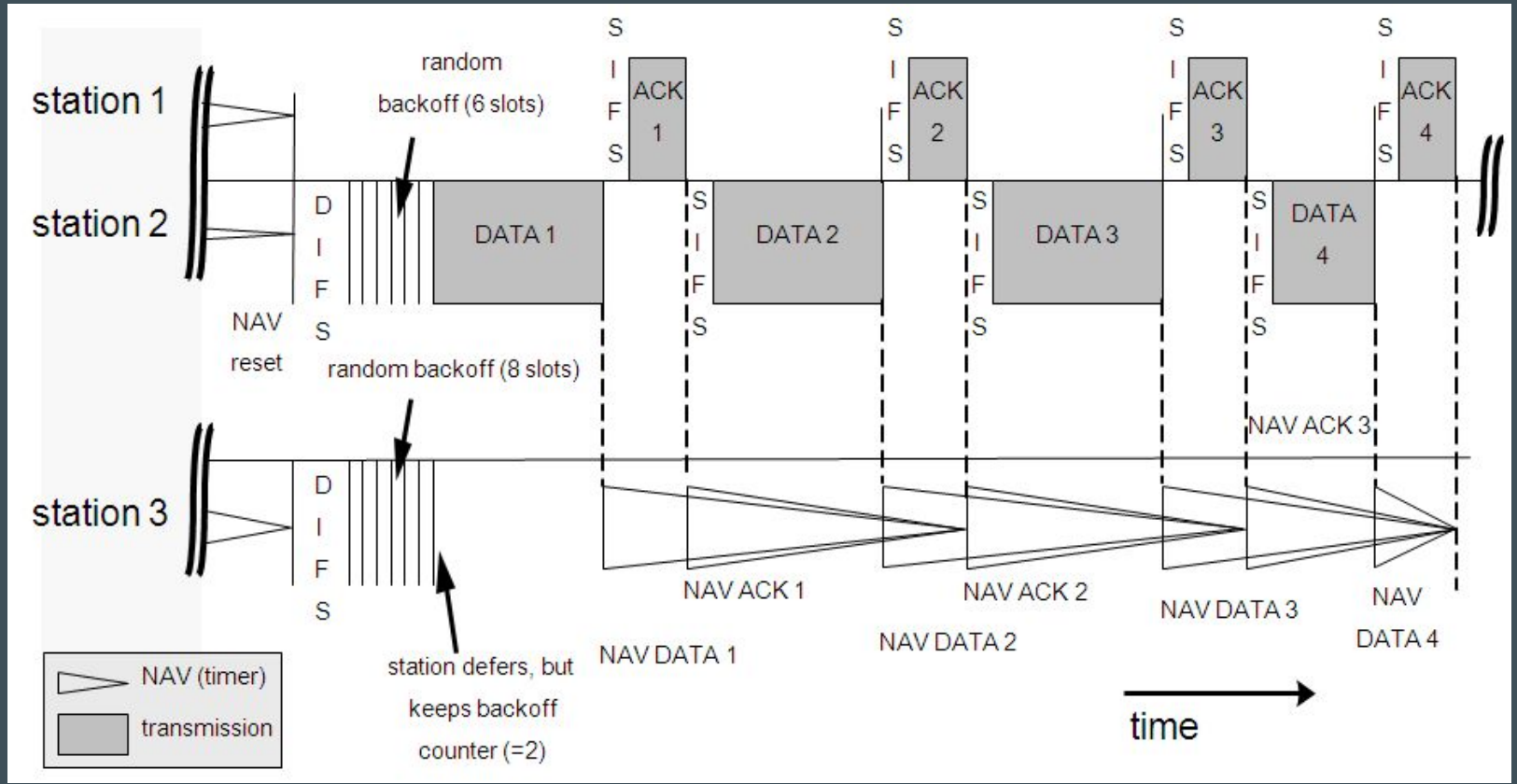

Jemula802 emulation results. (file:/C:/cygwin64/home/smangold/Source/Eclipse/jemula802/scenarios/Examples/4stations.xml)
30ms
40ms
50ms
60ms
70ms
80ms
Station 1
Station 2
Station 3
Station 4


Jemula802 emulation results. (file:/C:/cygwin64/home/smangold/Source/Eclipse/jemula802/scenarios/Examples/4stations.xml)
12.28ms
12.38ms
12.48ms
12.58ms
Station 1
Station 2
Station 3
Station 4
12.437 ms
6Mb/s
DA:2
ack 2
12.481 ms
12.421 ms
6 (15) 12.533 ms
12.587 ms
6Mb/s
DA:1
data 2

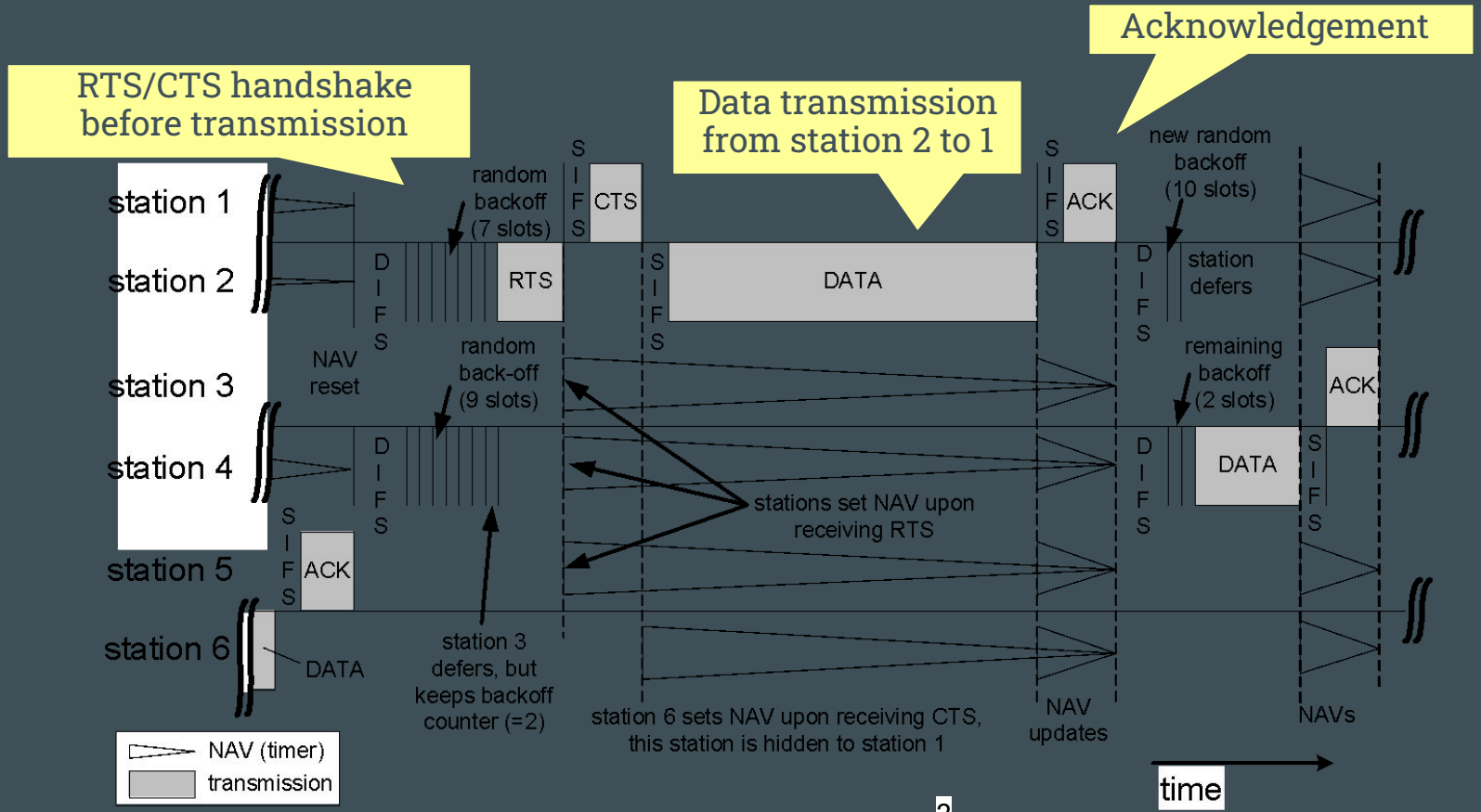
802.11 Fragmentation

- Data frames (MSDUs) transmitted in more than one MPDU
  - If the MSDU length exceeds a certain threshold
  - To reduce duration of collision times
- Each MPDU is acknowledged individually
  - This results in overhead (see also block-ACK of 802.11e/n)
- MPDUs protect subsequent transmissions of their ACK responses and the following data frame with NAV
- Defragmentation
  - Process of recombining MPDUs into a single MSDU at the intended receiver

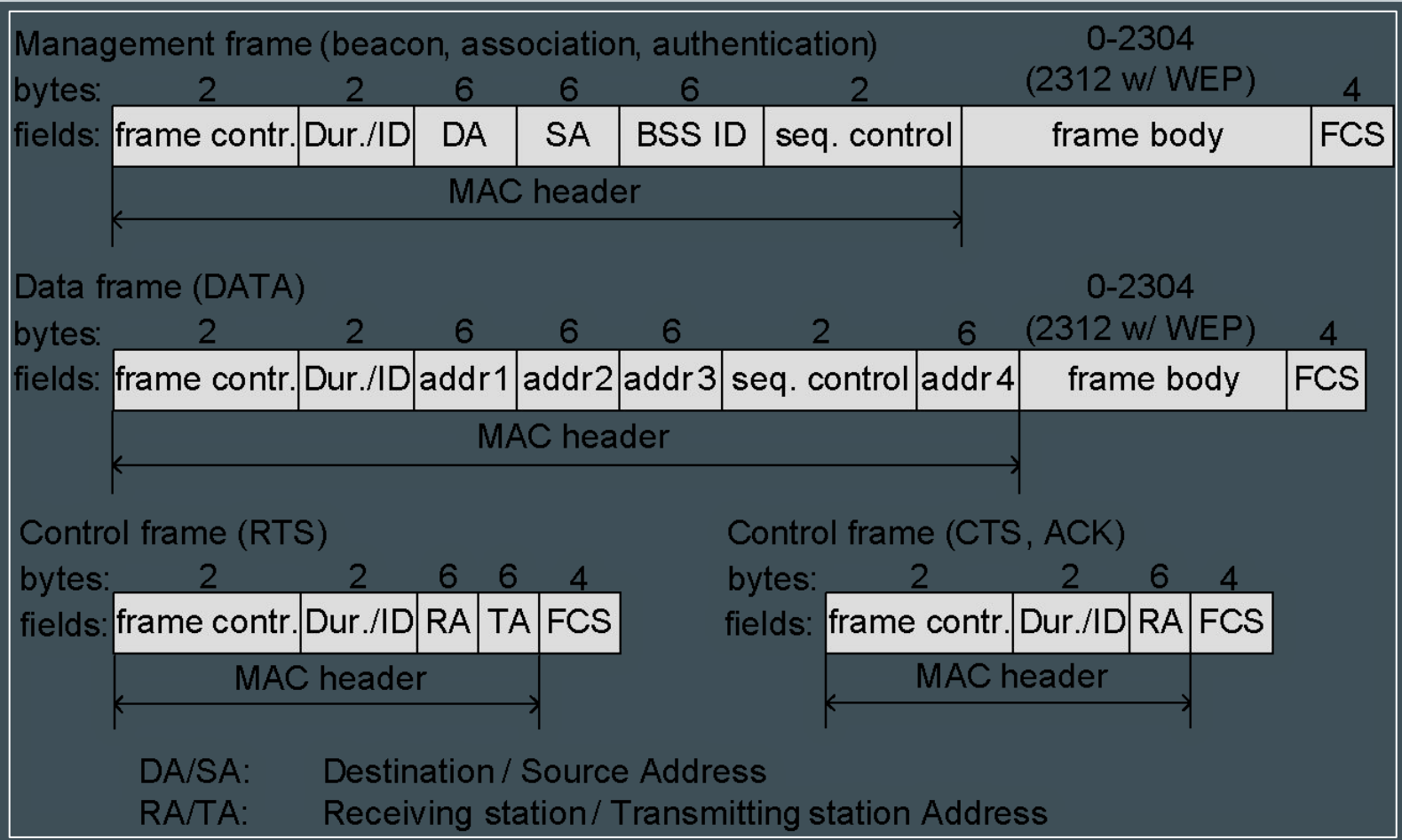
802.11 Fragmentation Illustrated



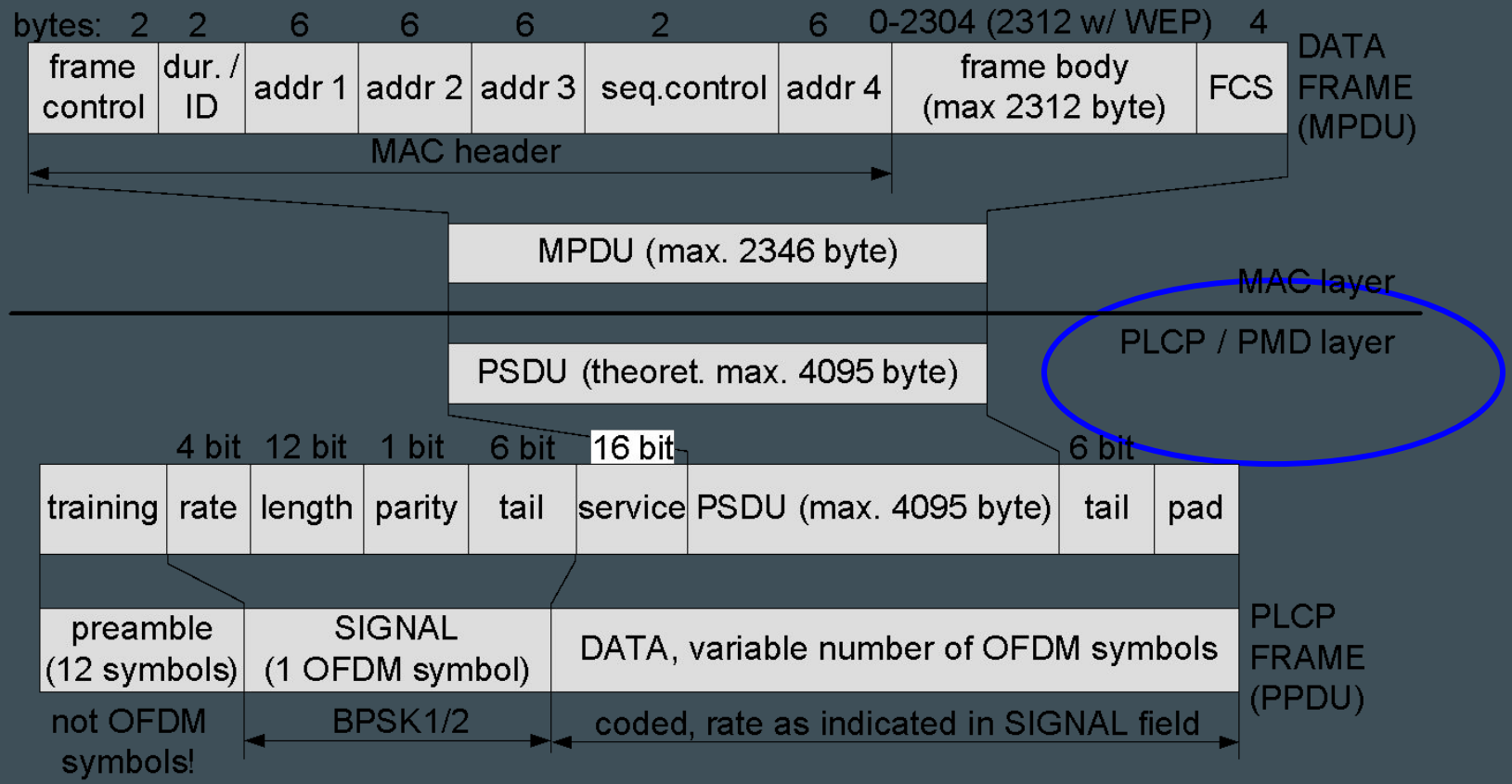
Hidden Stations and RTS / CTS



MAC Frames (Mac Service Data Units, MSDUs)



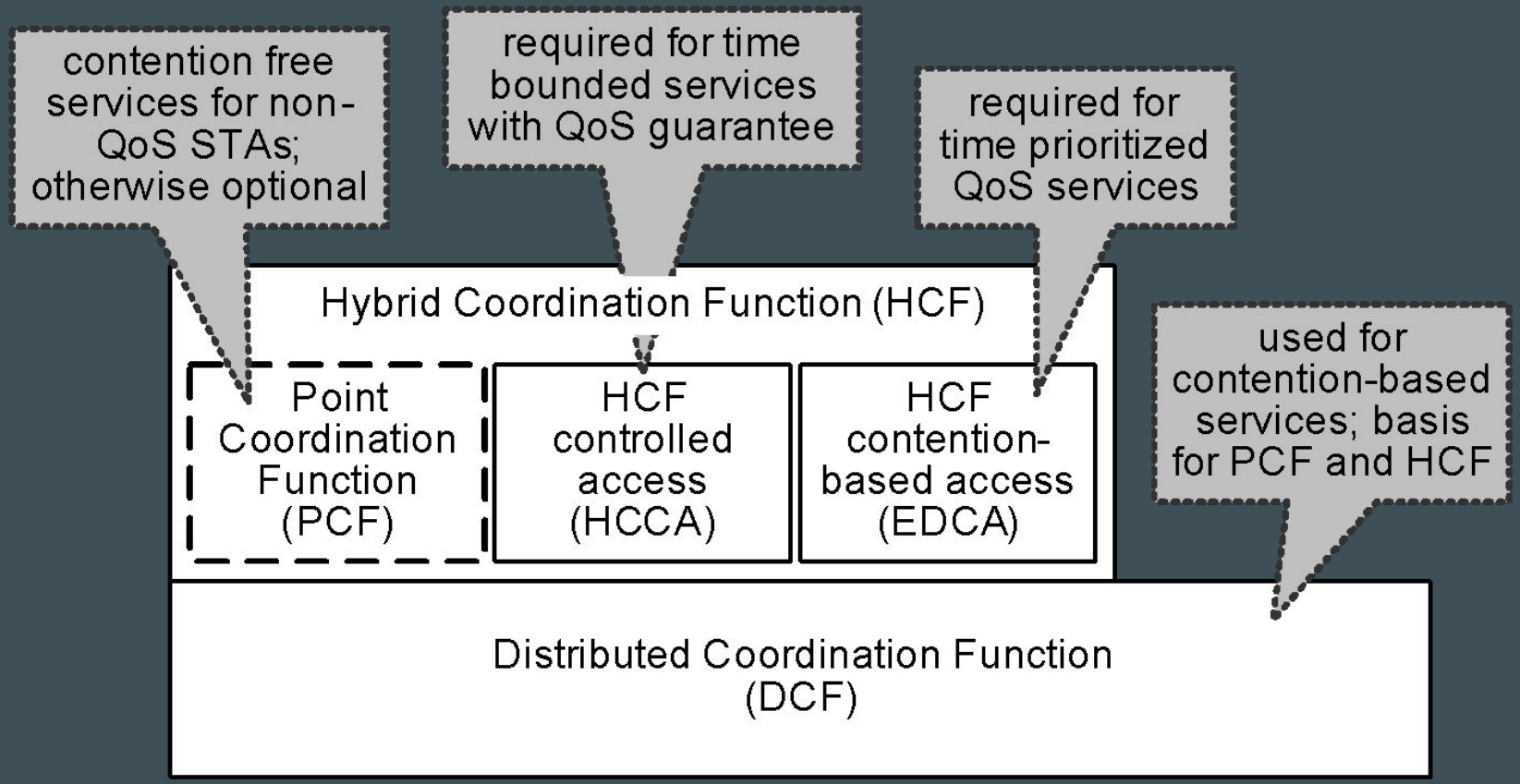
802.11 Frame Mapping MAC <-> PHY



IEEE 802.11e MAC Enhancements

- Layer-2 MAC protocol with backward compatibility
- Increasing demand for Quality-of-Service (QoS) was motivation for extending this standard
  - IEEE 802.11e standard was approved in 2005
  - Intel PRO/Wireless 3945ABG chipset supports 802.11e for VoIP and video streaming
- Hybrid Coordination Function (HCF) provides QoS
- HCF defines two new coordination functions
  - Controlled channel access HCCA
  - Contention-based channel access EDCA

MAC Enhancements Illustrated



802.11e Terminology

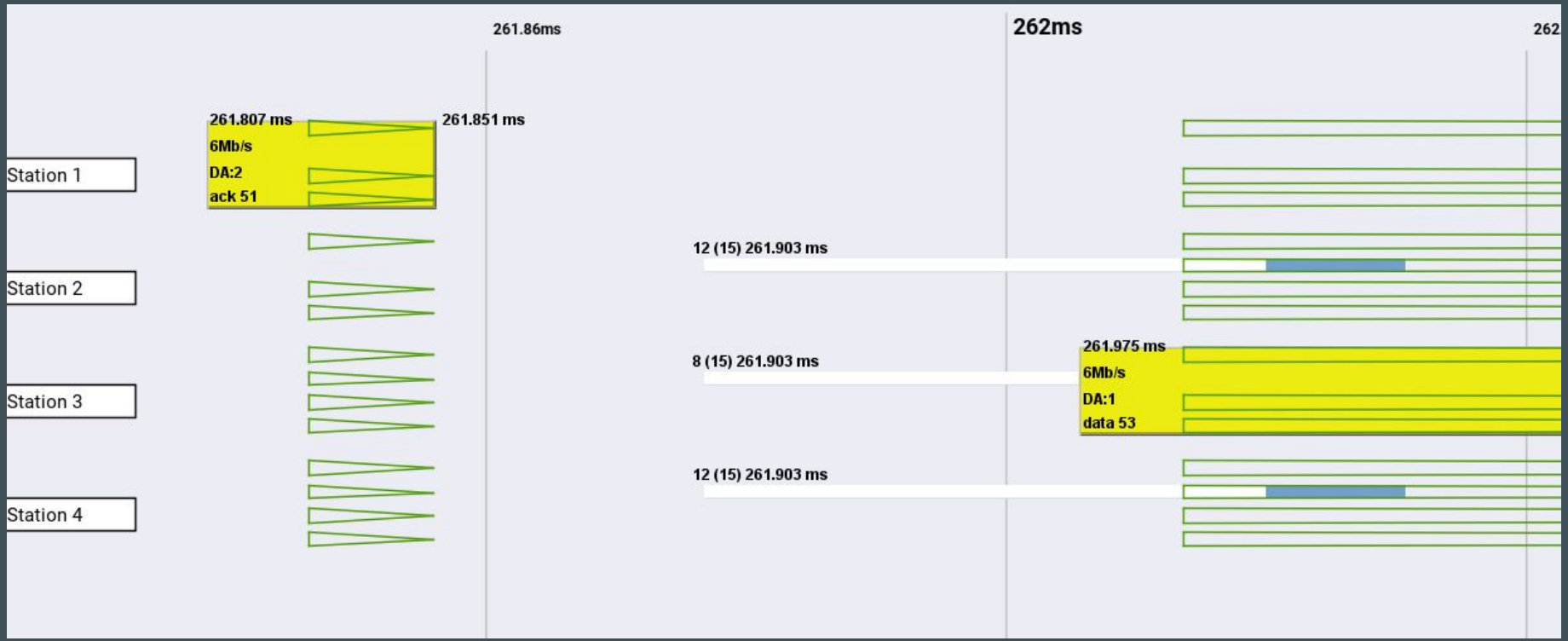
- Quality-of-service supporting Basic Service Set (QBSS)
  - A BSS that includes an 802.11e-compliant Hybrid Coordinator (HC)
- Transmission Opportunity (TXOP)
  - Defined by its starting time and duration
  - Obtained by 802.11e station (more precisely by a backoff entity)
  - TXOP obtained via contention-based medium access, is EDCA-TXOP
  - EDCA-TXOP duration is limited by TXOPlimit
  - TXOP obtained by the HC via the controlled medium access is HCCA-TXOP
  - HCCA-TXOP is not limited by TXOPlimit
- Controlled Access Phase (CAP)
  - Time period in which the HC has control over the medium (in CFP or in CP)

802.11e EDCA Priorities

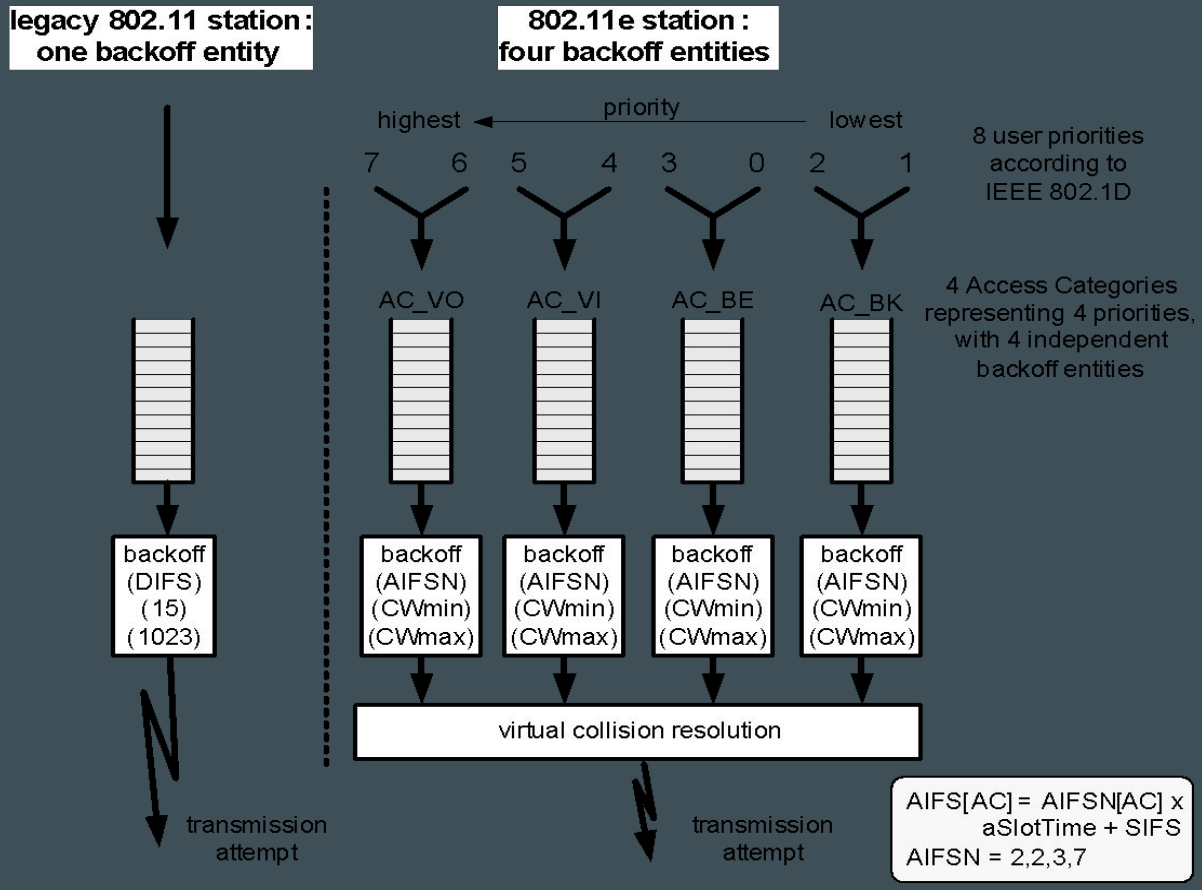
- EDCA enables a probability based support of “Quality of Service” by introducing four Access Categories (ACs)
  - each AC has different parameter sets of contention-based access
- Inside 802.11e station: ACs are implemented by parallel backoff entities with a so-called “virtual” collision resolution
- Channel access after medium sensed as idle
  - For certain time duration (AIFS instead of DIFS)
  - And after random waiting time drawn from CW

Four Access Categories (Backoff Entities, Priors)

- Four Backoff Entities in XML scenarios (EDCA default)



802.11e EDCA Priorities Illustrated

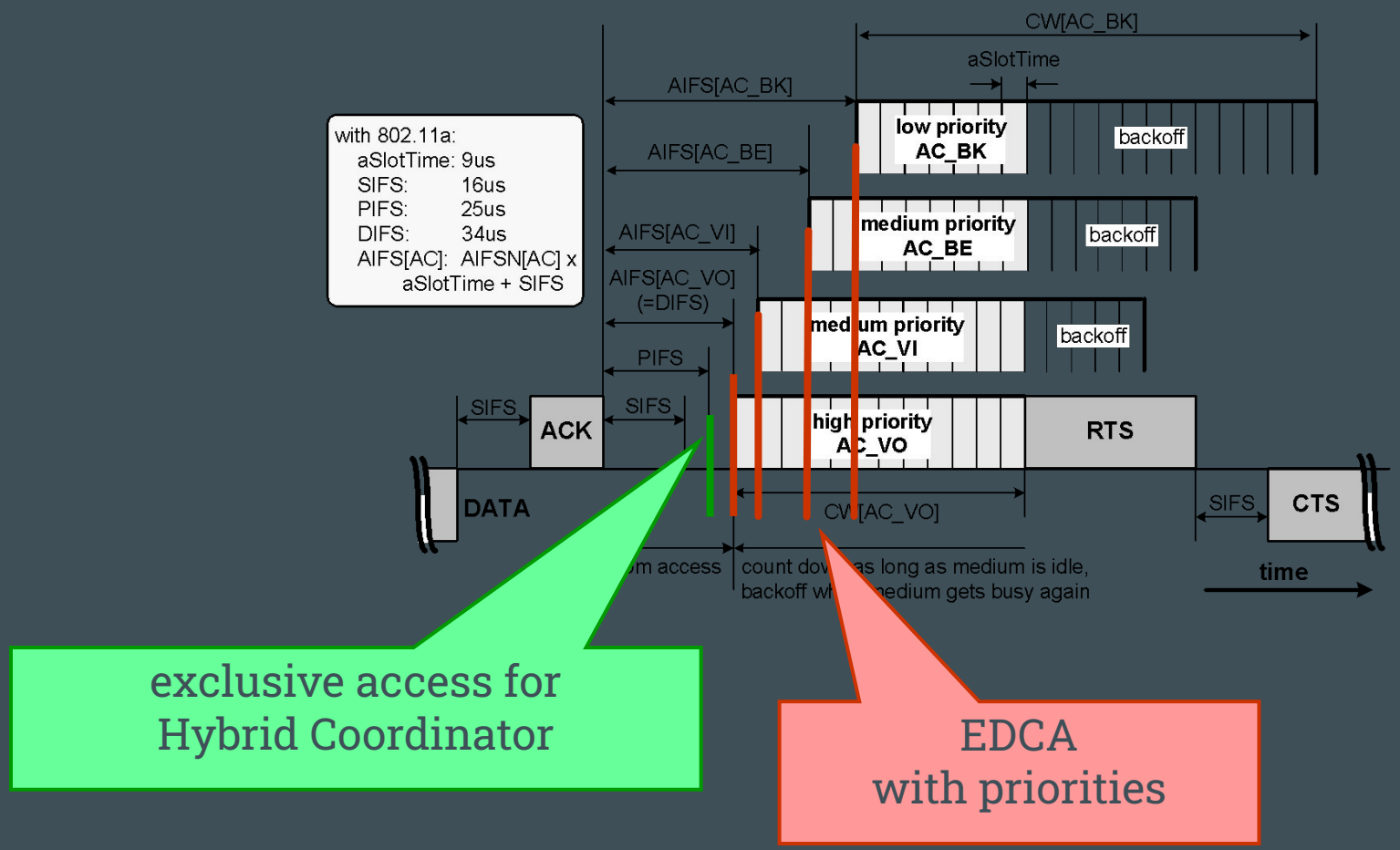


QoS Support with 802.11e EDCA

- Four Access Categories (ACs)
  - Background (AC_BK), Best Effort (AC_BE)
  - Video (AC_VI) and Voice (AC_VO)
- EDCA prioritization with parameters
  - AIFS[AC] in microseconds equivalent to AIFSN [AC]
  - CWmin[AC]
  - CWmax[AC]

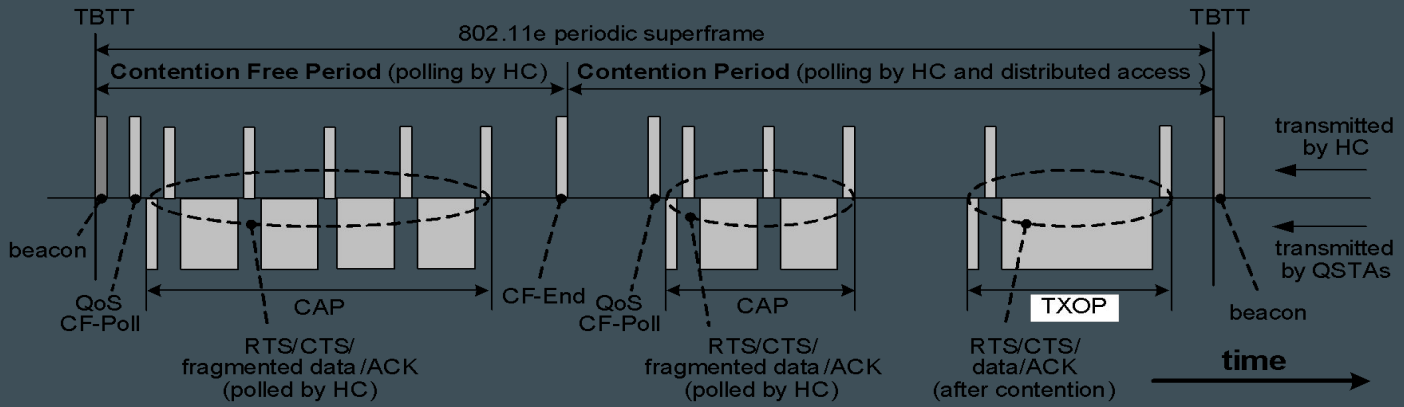
AC_BK	15	1023	7	79 us
				43 us
AC_VI	7	15	2	34 us
				34 us

QoS Support Illustrated

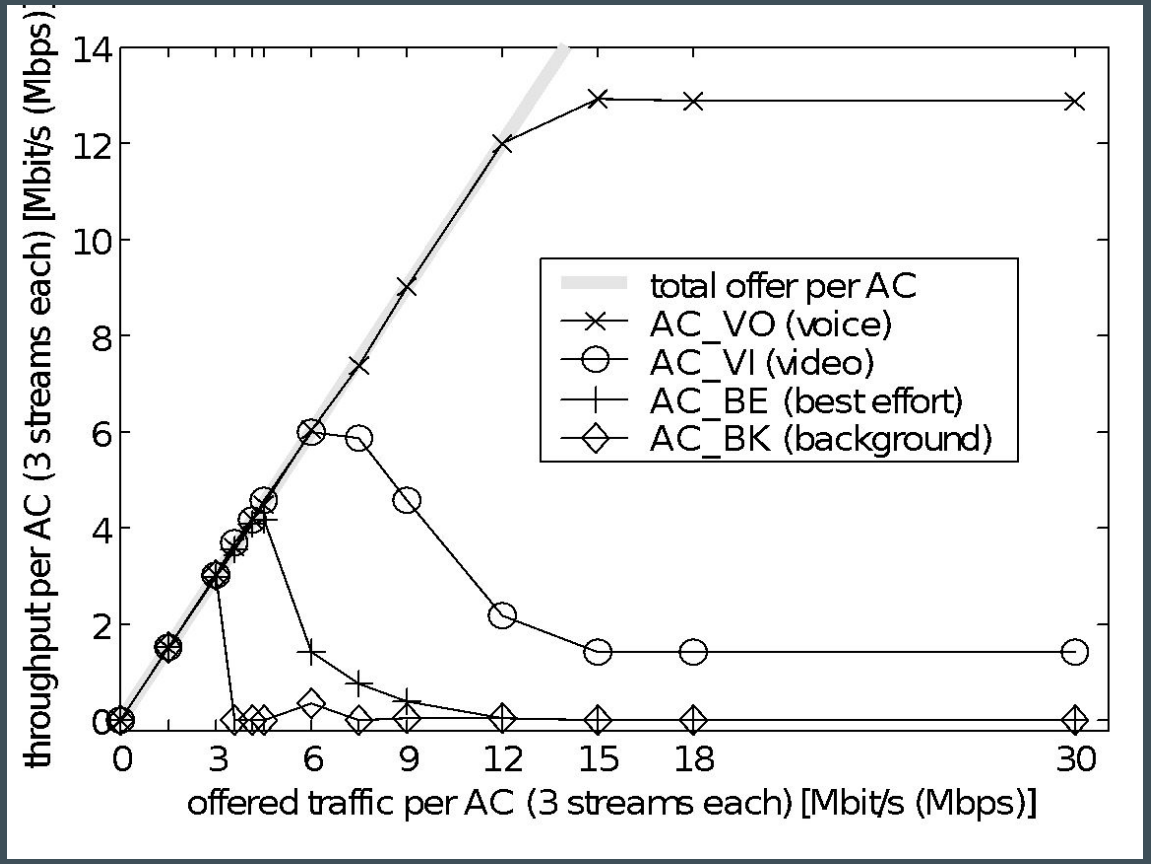


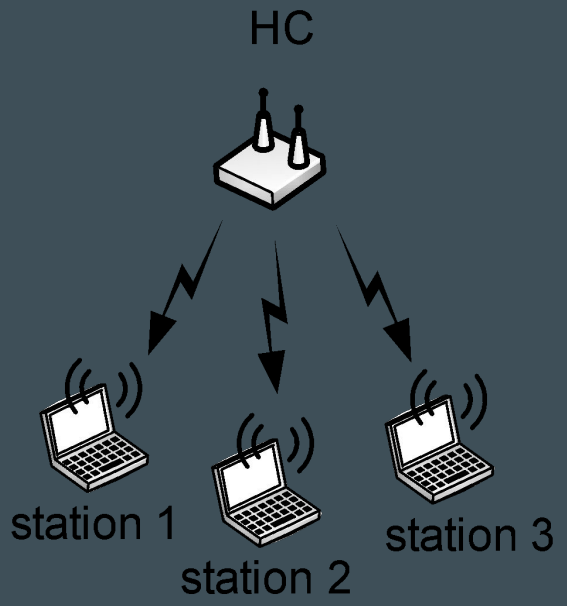
802.11e MAC Periodic Superframe Period

- Centralized operation
  - HCF Controlled Access (HCCA), exclusive access in “Contention Free Period”
- Decentralized, contention-based operation
  - Enhanced Distributed Channel Access (EDCA)
  - Listen-before-talk and backoff in “Contention Period”

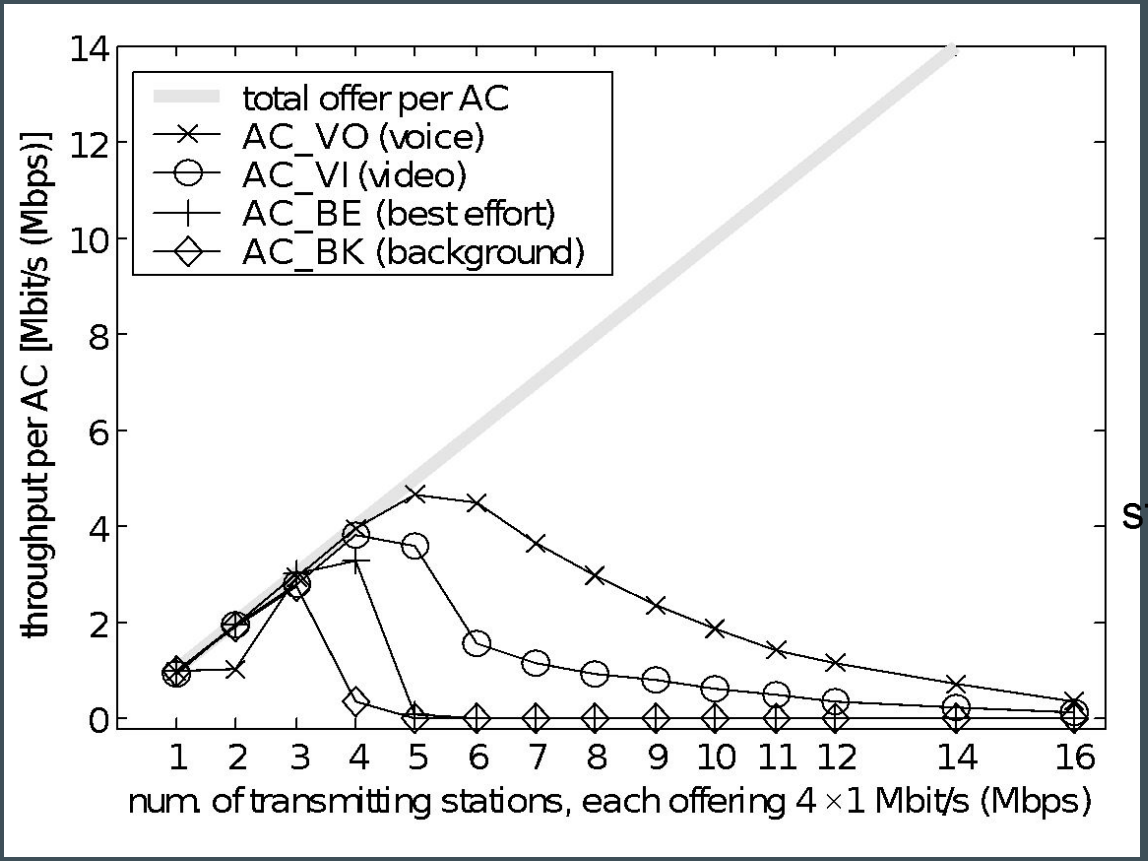


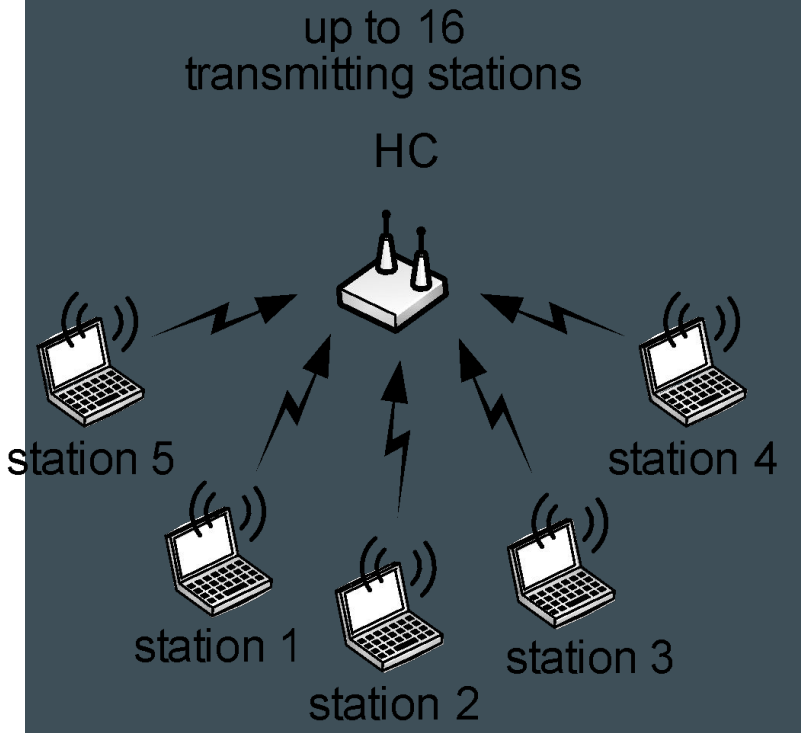
EDCA Throughput (Simulation 1/2)





EDCA Throughput (Simulation 2/2)

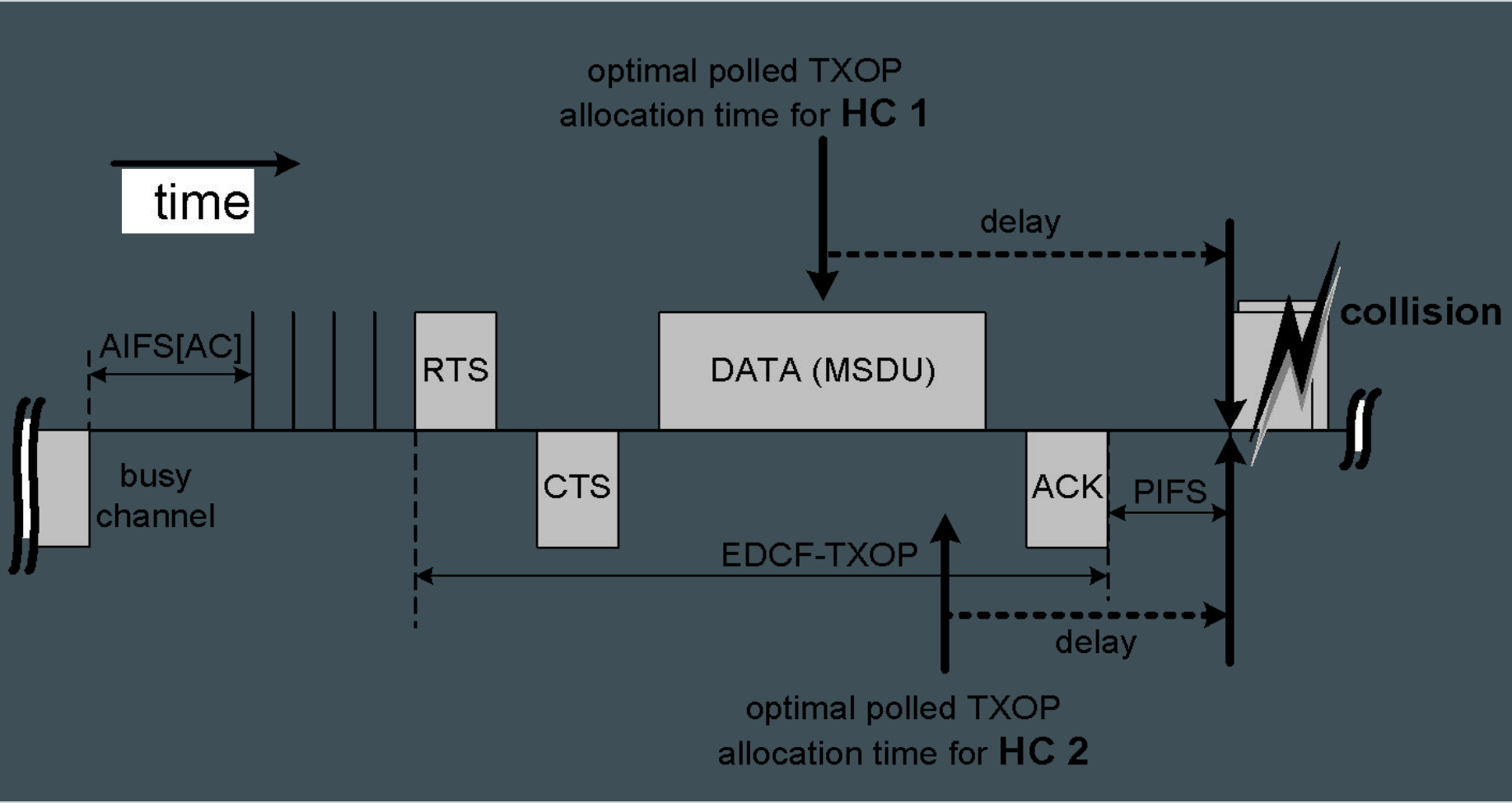




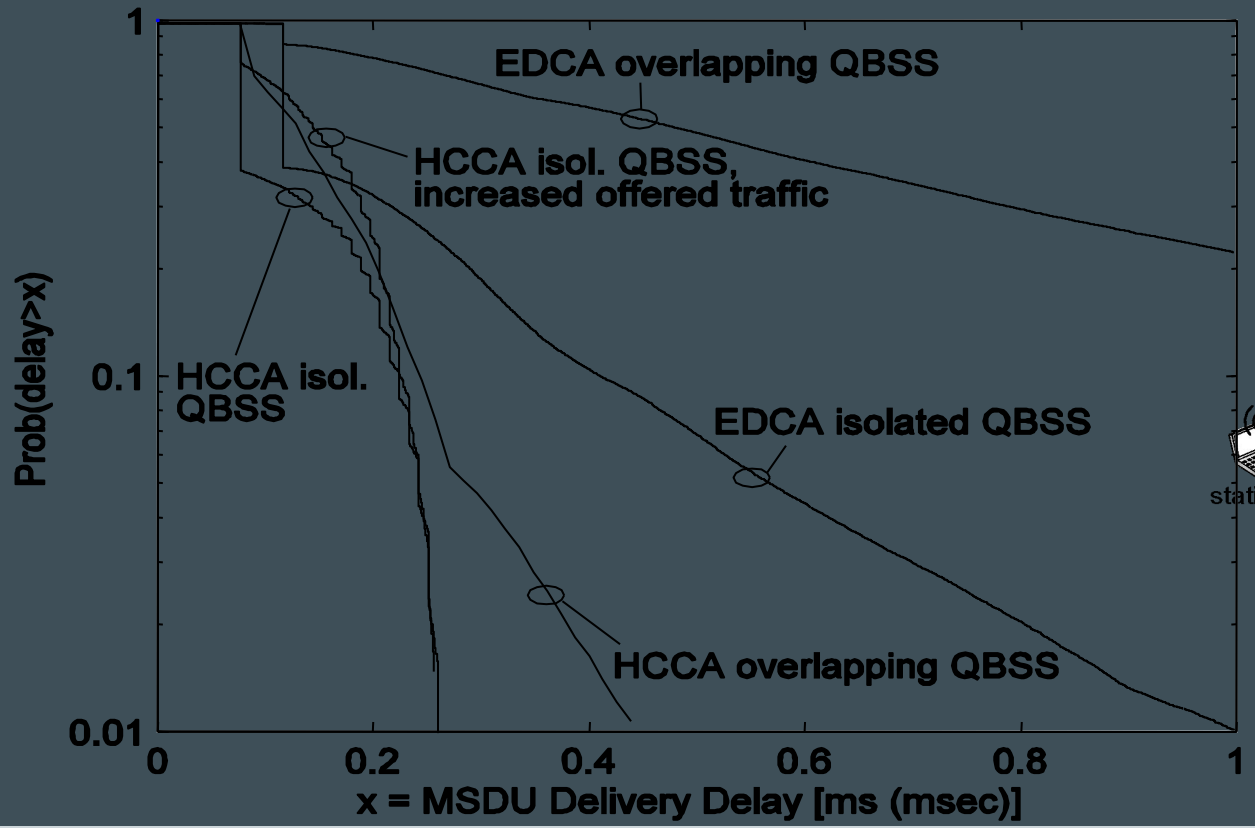
The Problem Of Overlapping BSSs

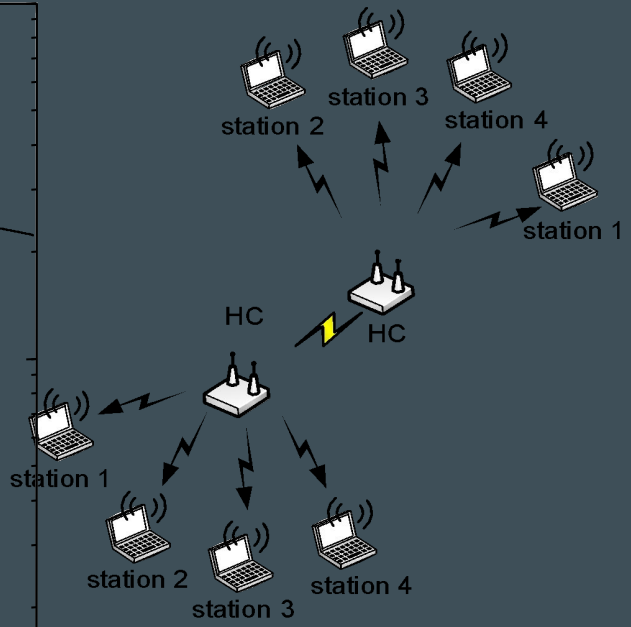
- Isolated BSS
  - Unpredictable delays may occur
  - However, maximum delay under control by HC, with the help of EDCA-TXOPlength
- Overlapping BSSs with more than one HC
  - HC has no control over opponent HC's TXOPs
  - QoS CF-Polls may collide, if both HCs access simultaneously the medium
- This is an unsolved coexistence problem

Overlapping BSSs Illustrated



Overlapping BSS Delays (Simulation)





IEEE 802.11e Summary

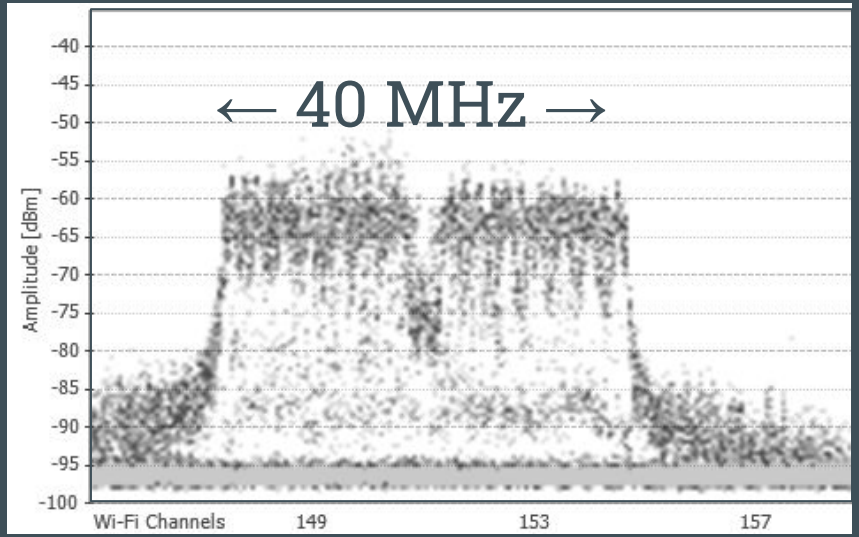
- 802.11e HCCA introduces centralized control for QoS guarantees
- Contention-based medium access of 802.11e EDCA implies limited capability to support QoS
- Number of contending backoff entities determines (1) the level of supported QoS and (2) differentiability of QoS
- The overlapping BSS problem is not addressed
  - Dynamic Frequency Selection (DFS) helps to mitigate this problem

IEEE 802.11n: High Throughput (HT)

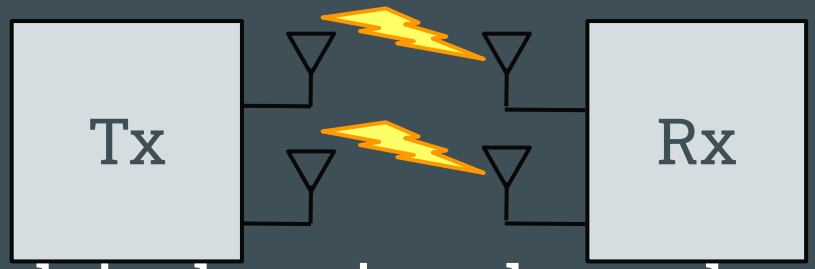
- PHY features
  - PPDU formats for 20/40 MHz operation
  - MIMO operation (spatial multiplexing), Tx beamforming, antenna selection
- MAC features
  - Reduced Interframe Space (RIFS < SIFS)
  - MSDU/MPDU aggregation, block acknowledgement (see 802.11e)
- Throughput increase
  - 2x2 antenna operation at 20 MHz → 100 Mb/s
  - 4x4 antenna operation at 40 MHz → 600 Mb/s

802.11n Channel Bundling

- Pair of 20 MHz channels
- Implementation
  - AP defines primary and secondary 20 MHz channel
  - Primary channel is used for coordination (beacon broadcast etc.)
  - Stations do not need secondary CCA
  - Increasing collision probability with overlapping BSS on secondary channel
- 40 MHz transmissions may be interfered on secondary channel

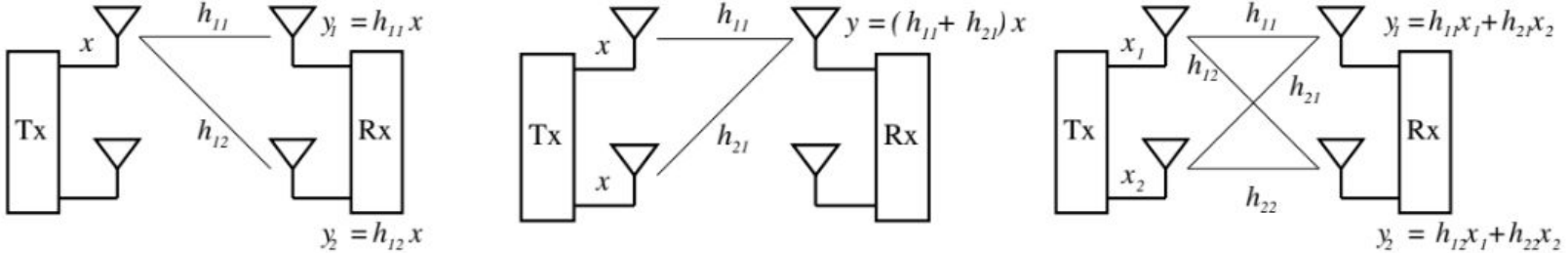


Multiple Input Multiple Output (MIMO) 1/3



- Diversity: exploits location-dependent independent fading
- TX Beamforming: Increases signal gain (same for RX)
  - Same signal is emitted from each transmit antenna with appropriate phase shift such that the signal power is maximized at receiver
  - Requires knowledge of channel state information (CSI) at transmitter
- Spatial multiplexing: create true orthogonal channels
  - Separately encoded data signals

Multiple Input Multiple Output (MIMO) 2/3

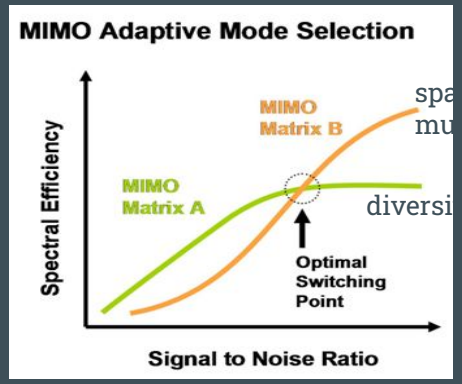


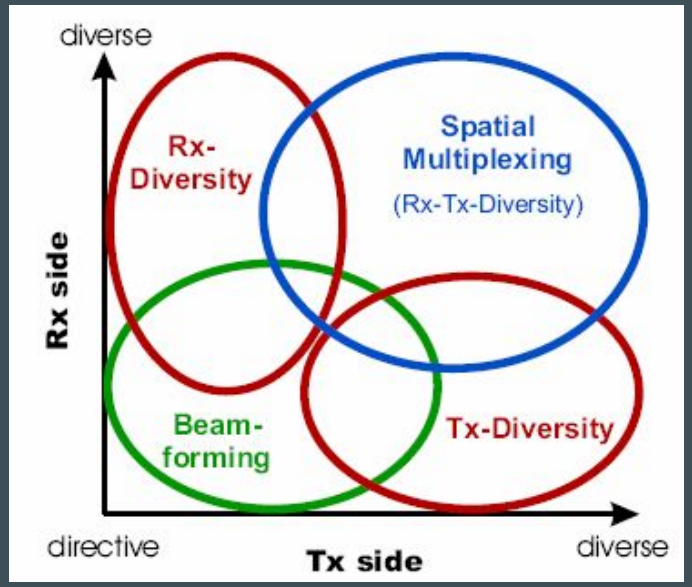
(a) Receive diversity (b) Transmit diversity (c) Spatial multiplexing

Figure 2: Using some of the transmit/receive antennas in an example 2x2 system to exploit diversity and multiplexing gain. x_i and y_i represent transmitted and received signals. The channel gain h_{ij} is a complex number indicating a signal's attenuated amplitude and phase shift over the channel between the ith transmit antenna and the jth receive antenna. The received signals y_i will additionally include thermal RF noise.

Multiple Input Multiple Output (MIMO) 3/3

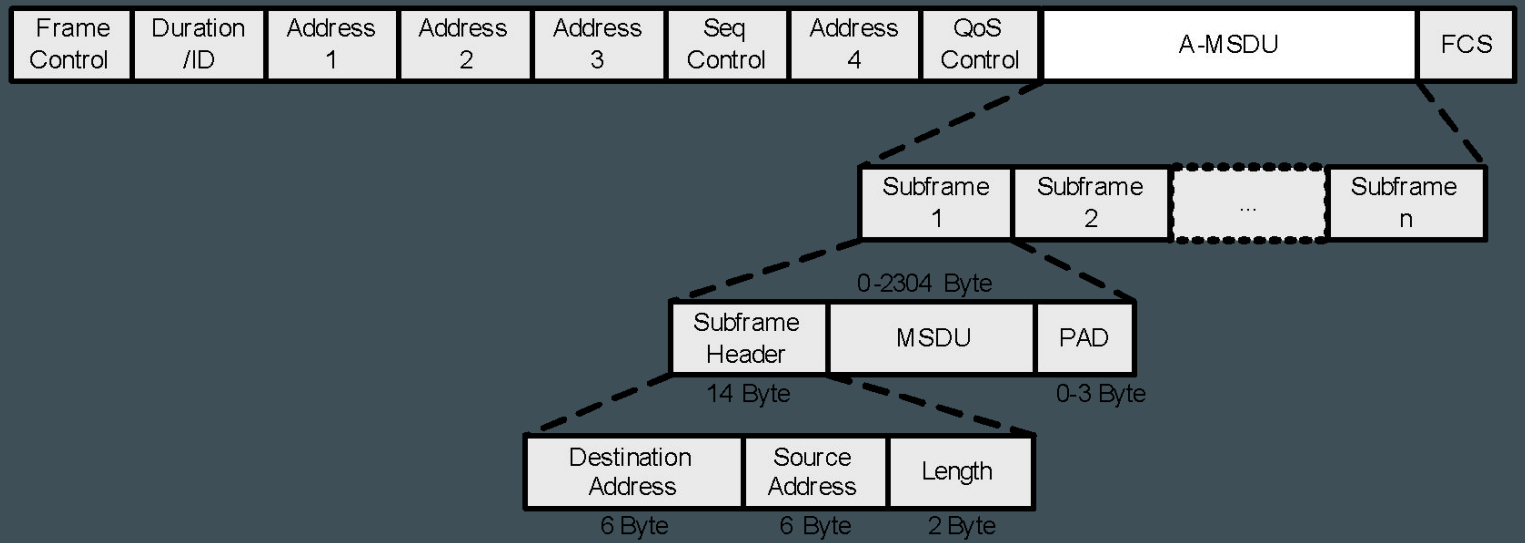
- Spatial multiplexing + beamforming
  - If channel is known at transmitter
- Spatial multiplexing + diversity coding
  - If decoding reliability is in trade-off
- Challenge: λ/2 antenna spacing
  - Requires large antenna arrays





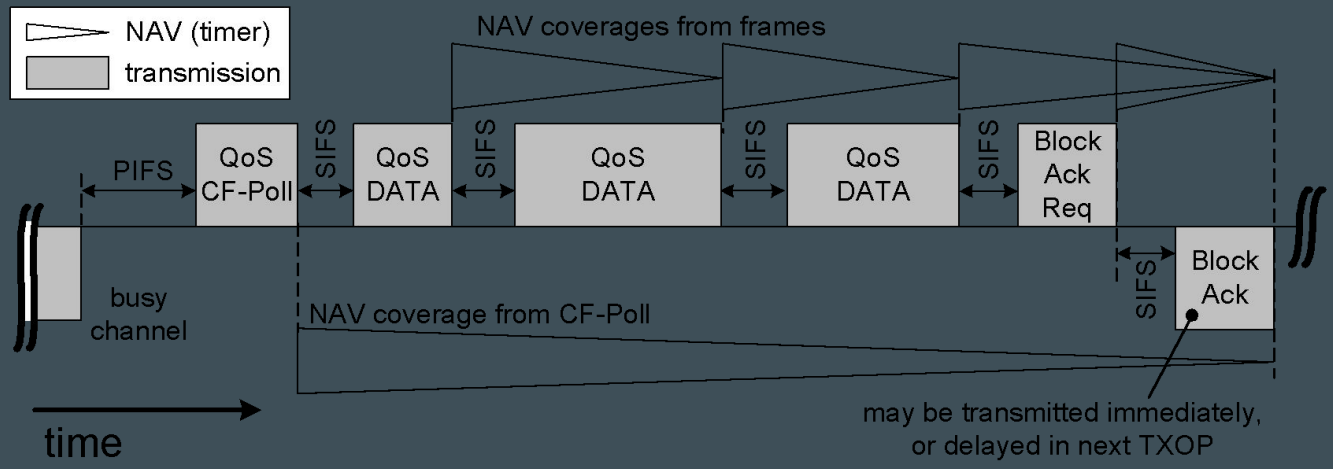
802.11n MSDU Aggregation

- Aggregation of multiple frames
  - Reduce protocol overhead
  - A-MSDU is aggregated with multiple MSDUs
  - Aggregated MSDU frames need to have same address type & priority class



802.11e/n Block Acknowledgement

- 802.11e and 802.11n extensions
  - Implicit block acknowledgement request (BAR) → reduced overhead
  - Compressed block acknowledgement → reduced bitmap size
  - Multi-TID (Traffic Identifier) block acknowledgement → reduced overhead
  - Immediate and delayed Block ACK depending on traffic type



IEEE 802.11n Summary

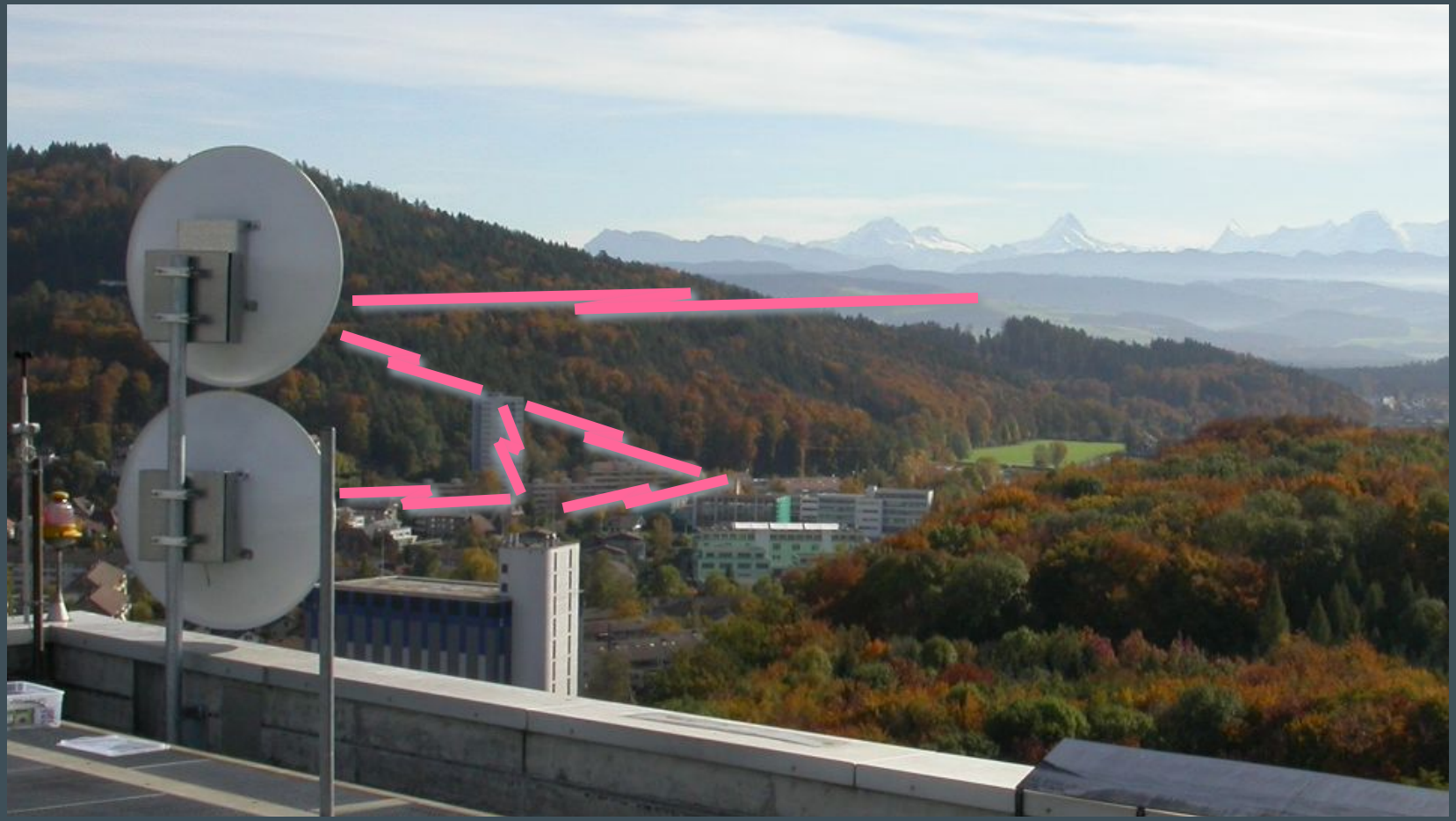
- Throughput performance is significantly increased
- High increase in indoor scenarios
  - Ideal environment for MIMO
- Smaller performance increase in outdoor scenarios
  - Mainly due to channel bundling
- Good complement to 11e/11s: QoS & mesh & high rates
  & coverage

WLAN Long Range Links (~ 20 km)





Wi-Fi Mesh



Potential Benefits Of Mesh Networks

- No need for infrastructure
  - Self configuration, easy deployment, cost efficiency
- Robustness against network failures or damages
  - System repairs a break-down by itself, allowing higher availability
- System adapts to changes
- Scalability: self-organizing networks enable scalability

IEEE 802.11s: Overview

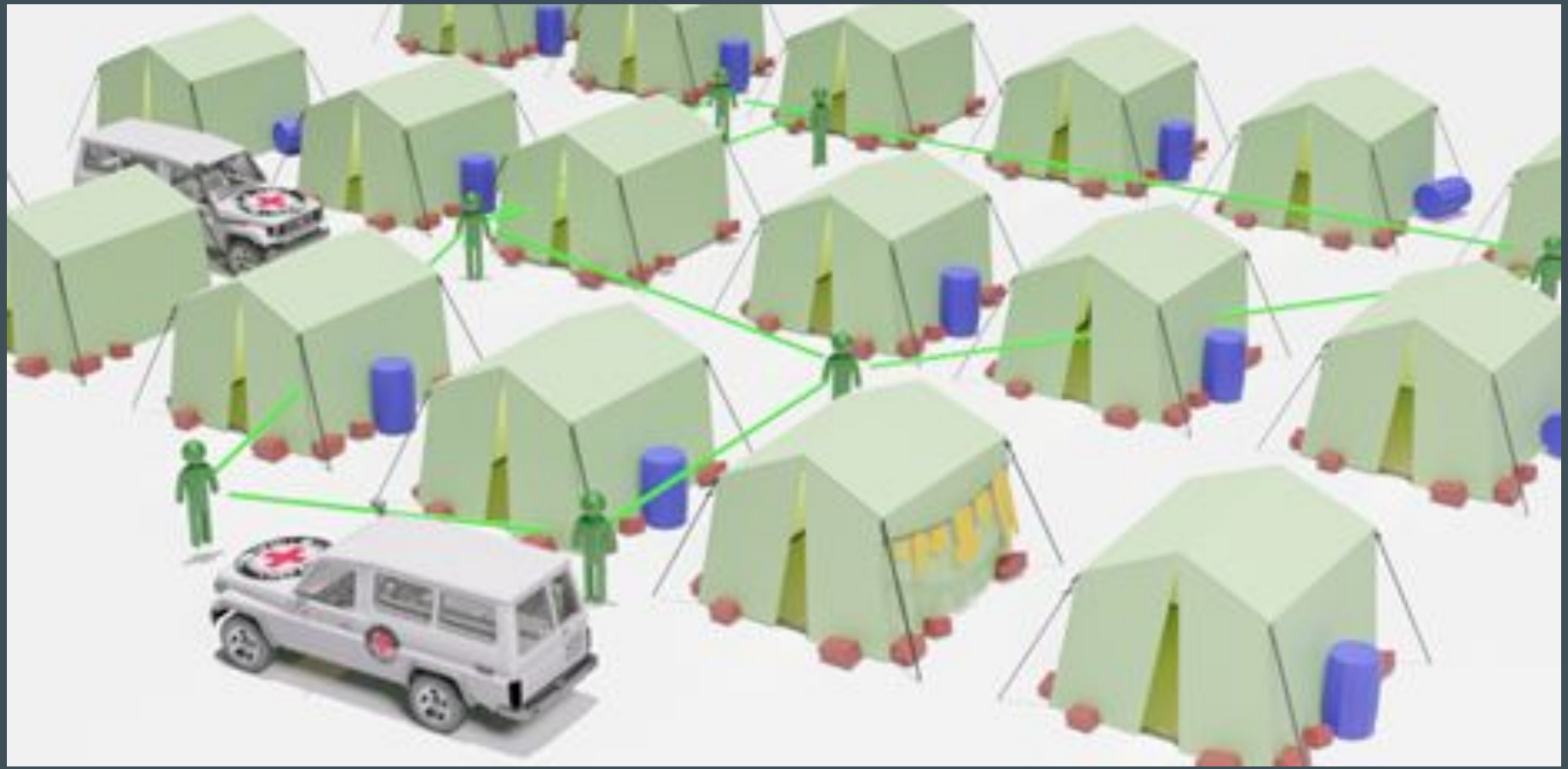
- Primary scope
  - Small/medium mesh networks (~32 forwarding nodes)
  - Self-configuring, multi-hop topologies
  - Dynamic, radio-aware path selection in mesh
  - Support uni-, multi- & broadcast traffic
- Deployment usage scenarios
  - Military: battlefield communication
  - Public safety: emergency & disaster area communication
  - Residential: consumer electronics
  - Public access: operated networks
  - Office: enterprise & business networks

One Laptop per Child (OLPC)

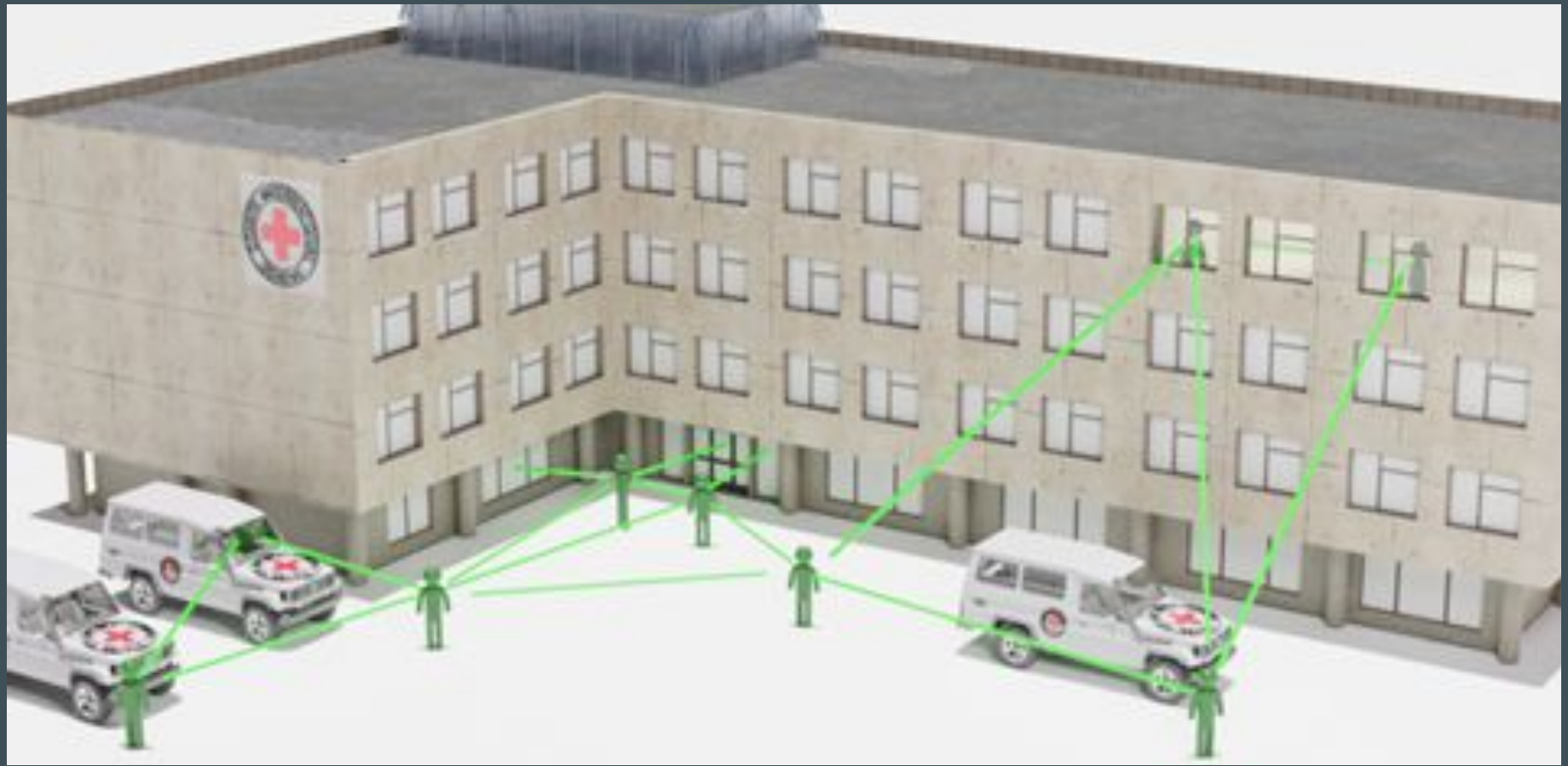
- IEEE 802.11s implementation
- Open source
- Mesh code partially implements 802.11s-D1.0
  - HWMP without tree based routing, and AODV
- Operates simultaneously as Station and Mesh Point
- WLAN chipset independent of main CPU



Mesh Networks in Disaster Recovery (1 of 2)



Mesh Networks in Disaster Recovery (2 of 2)



Exercise: Wi-Fi Simulation with Jemula802

- Open in your course folder:
  02_Notebooks/03_Mobile_Systems/
  03.7_Wi-Fi_Simulation_with_Jemula802.ipynb
- Compile the simulator, run the example, and inspect the results.
- Complete Section 7.5, save the notebook, and upload it to Moodle.

